

Date: 17th March 2026

DAB-II

Section:A

Topic to be cover: Hypothesis testing of Proportion & comparison between 2 pop mean

Hypothesis testing of proportion

There are 6 basic steps of hypothesis testing for population mean:

- 1) State the claim & opposite to the claim
- 2) Level of Significance
- 3) Test Statistics
- 4) Critical region
- 5) Calculation
- 6) Conclusion

Roadmap of proportion testing

Hypothesis	Test Statistics	Critical region
$H_1: P > a$	$z = \frac{\hat{p} - P}{\sqrt{\frac{PQ}{n}}}$	$Z > Z_\alpha$ If critical region satisfies reject H_0
$H_1: P < a$		$Z < -Z_\alpha$ If critical region satisfies reject H_0
$H_1: P \neq a$		$Z > Z_{\alpha/2}$ or $Z < -Z_{\alpha/2}$ If critical region satisfies reject H_0

Exercise: 8.4 Elementary Statistics by Bluman

Question#5 **Home Ownership** A recent survey found that 68.6% of the population own their homes. In a random sample of 150 heads of households, 92 responded that they owned their homes. At the $\alpha = 0.01$ level of significance, does that suggest a difference from the national proportion?

Step#1

$H_0: P = 0.686$

$H_1: P \neq 0.686$

Step#2

Level of significance: $\alpha = 0.01$

Step#3

Test Statistics

$$z = \frac{\hat{p} - P}{\sqrt{\frac{PQ}{n}}}$$

Step#4

Critical region

$$Z > Z_{\alpha/2} \text{ or } Z < -Z_{\alpha/2}$$

$$Z > Z_{0.005} \text{ or } Z < -Z_{0.005}$$

$$Z > 2.576 \text{ or } Z < -2.576$$

If critical region satisfies reject H_0

Step#5

Calculation:

$$P = 0.686, Q = 0.314, \hat{p} = \frac{92}{150} = 0.613, n = 150,$$

$$z = \frac{\hat{p} - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.613 - 0.686}{\sqrt{\frac{0.686 * 0.314}{150}}} = -1.9263$$

$-1.9263 > 2.576$ or $-1.9263 < -2.576 \rightarrow$ doesn't satisfy

Step#6

Conclusion: As CR doesn't satisfy therefore there is no evidence to reject H_0 at 0.01 level of significance we conclude that owned house proportion is equal to 0.686

Question# 7. Overweight Children Health issues due to being overweight affect all age groups. Of children and adolescents 6–11 years of age, 18.8% are found to be overweight. A school district randomly sampled 130 in this age group and found that 20 were considered overweight. At $\alpha = 0.05$ is this less than the national proportion?

Step#1

$$H_0: P \geq 0.188$$

$$H_1: P < 0.188$$

Step#2

Level of significance: $\alpha = 0.05$

Step#3

Test Statistics

$$z = \frac{\hat{p} - P}{\sqrt{\frac{PQ}{n}}}$$

Step#4

Critical region

$$Z < -Z_{\alpha}$$

$$Z < -Z_{0.05}$$

$$Z < -1.645$$

If critical region satisfies reject H_0

Step#5

Calculation:

$$P = 0.188, Q = 0.812, \hat{p} = \frac{20}{130} = 0.154, n = 130,$$

$$z = \frac{\hat{p} - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.154 - 0.188}{\sqrt{\frac{0.188 * 0.812}{130}}} = -0.99218$$

$-0.992 < -1.645 \rightarrow$ doesn't satisfy

Step#6

Conclusion: As CR doesn't satisfy therefore there is no evidence to reject H_0 at 0.05 level of significance we conclude that overweight proportion is greater & equal to 0.188.

Question#19: **Football Injuries** A report by the NCAA states that 57.6% of football injuries occur during practices. A head trainer claims that this is too high for his conference, so he randomly selects 36 injuries and finds that 17 occurred during practices. Is his claim correct, at $\alpha = 0.05$?

Step#1

$$H_0: P \geq 0.576$$

$H_1: P < 0.576$

Step#2

Level of significance: $\alpha = 0.05$

Step#3

Test Statistics

$$z = \frac{\hat{p} - P}{\sqrt{\frac{PQ}{n}}}$$

Step#4

Critical region

$$Z < -Z_{\alpha}$$

$$Z < -Z_{0.05}$$

$$Z < -1.645$$

If critical region satisfies reject H_0

Step#5

Calculation:

$$P = 0.576, Q = 0.424, \hat{p} = \frac{17}{36} = 0.4722, n = 36$$

$$z = \frac{\hat{p} - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.4722 - 0.576}{\sqrt{\frac{0.576 * 0.424}{36}}} = -1.26$$

$-1.26 < -1.645 \rightarrow$ doesn't satisfy

Step#6

Conclusion: As Cr doesn't satisfy therefore there is no evidence to reject H_0 at 0.05 level of significance we conclude that injuries proportion is greater & equal to 57.6%.

Statistical Inference for Comparison Between Two Means (σ_1^2 and σ_2^2 Known):

In Statistical Inference, comparing two population means helps researchers or financial analysts determine whether differences between groups are statistically significant when population variances are known. In finance, this method can be used to compare the average returns of two

investment portfolios, the mean daily returns of two stocks, or the average interest rates offered by two banks. By analyzing sample data from each group, statistical inference allows us to estimate and test the difference between the population means ($\mu_1 - \mu_2$).

Point Estimate:

The point estimate of the difference between the two population means is the difference between the sample means:

$$\mu_1 - \mu_2 = \bar{X}_1 - \bar{X}_2$$

Interval Estimate (Confidence Interval):

A confidence interval for the difference between the two means is given by:

$$(\bar{X}_1 - \bar{X}_2) \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Where : $\bar{X}_1$ is sample mean from population 1, $\bar{X}_2$ is sample mean from population 2, n_1 is sample size of population 1, n_2 is sample size of population 2, σ_1^2 is variance of population 1, σ_2^2 is variance of population 2, $Z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$ is margin of error.

Exercise 9.1, Elementary statistics by Bluman.

Question#17. Problem-Solving Ability Two groups of students are given a problem-solving test, and the results are compared. Find the 90% confidence interval of the true difference in means.

Mathematics majors	Computer science majors
$\bar{X}_1 = 83.6$	$\bar{X}_2 = 79.2$
$\sigma_1 = 4.3$	$\sigma_2 = 3.8$
$n_1 = 36$	$n_2 = 36$

$$CL=90\%, \alpha/2=5\%= 0.05 \quad Z_{0.05}=1.645$$

$$(\bar{X}_1 - \bar{X}_2) \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \rightarrow (83.6 - 79.2) \pm 1.645 \sqrt{\frac{4.3^2}{36} + \frac{3.8^2}{36}} \rightarrow 2.8267 \text{ to } 5.973$$

We are 90% confident that population difference lies between 2.8267 to 5.973

Question# 20. Battery Voltage Two brands of batteries are tested, and their voltage is compared. The data follow. Find the 95% confidence interval of the true difference in the means. Assume that both variables are normally distributed.

Brand X	Brand Y
$\bar{X}_1 = 9.2$ volts	$\bar{X}_2 = 8.8$ volts
$\sigma_1 = 0.3$ volt	$\sigma_2 = 0.1$ volt
$n_1 = 27$	$n_2 = 30$

$$(\bar{X}_1 - \bar{X}_2) \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Hypothesis Testing:

To test a claim about the difference between two population means:

Test Statistic:

$$z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

The calculated z-value is compared with the critical value from the standard normal distribution to determine whether to reject the null hypothesis about the difference between the two population means.

Roadmap of Comparison between 2 mean when population variance is known

Hypothesis	Test Statistics	Critical region
$H_1: \mu_1 - \mu_2 > a$	$z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$	$Z > Z_{\alpha}$
$H_1: \mu_1 - \mu_2 < a$		$Z < -Z_{\alpha}$
$H_1: \mu_1 - \mu_2 \neq a$		$Z > Z_{\alpha/2}$ or $Z < -Z_{\alpha/2}$

Exercise 9.1, Elementary statistics by Bluman.

Question# 11. Women Science Majors In a study of women science majors, the following data were obtained on two groups, those who left their profession within a few months after graduation (leavers) and those who remained in their profession after they graduated (stayers). Test the claim that those who stayed had a higher science grade point average than those who left. Use $\alpha = 0.05$.

Leavers	Stayers
$\bar{X}_1 = 3.16$	$\bar{X}_2 = 3.28$
$\sigma_1 = 0.52$	$\sigma_2 = 0.46$
$n_1 = 103$	$n_2 = 225$

Step#1

Ho:

H₁:

Step#2

Level of significance: α

Step#3

Test Statistics

$$z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Step#4

Critical region

$$Z > Z_{\alpha}$$

$$Z < -Z_{\alpha}$$

$$Z > Z_{\alpha/2} \text{ or } Z < -Z_{\alpha/2}$$

If critical region satisfies reject Ho

Step#5

Calculation:

$$\bar{X}_1 =, \bar{X}_2 =, \sigma_1^2 =, \sigma_2^2 =, n_1 =, n_2 =$$

$$z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Step#6

Conclusion:

Question#10 Home Prices A real estate agent compares the selling prices of homes in two municipalities in southwestern Pennsylvania to see if there is a difference. The results of the study are shown. Is there enough evidence to reject the claim that the average cost of a home in both locations is the same? Use $\alpha = 0.01$.

Scott	Ligonier
$\bar{X}_1 = \$93,430^*$	$\bar{X}_2 = \$98,043^*$
$\sigma_1 = \$5602$	$\sigma_2 = \$4731$
$n_1 = 35$	$n_2 = 40$

*Based on information from RealSTATs.

Step#1

Ho:

H₁:

Step#2

Level of significance: α

Step#3

Test Statistics

$$z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Step#4

Critical region

$Z > Z_\alpha$

$$Z < -Z_{\alpha}$$

$$Z > Z_{\alpha/2} \text{ or } Z < -Z_{\alpha/2}$$

If critical region satisfies reject H_0

Step#5

Calculation:

$$\bar{X}_1 =, \bar{X}_2 =, \sigma_1^2 =, \sigma_2^2 =, n_1 =, n_2 =$$

$$z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Step#6

Conclusion:

Question# 9. Length of Hospital Stays The average length of “short hospital stays” for men is slightly longer than that for women, 5.2 days versus 4.5 days. A random sample of recent hospital stays for both men and women revealed the following. At $\alpha=0.01$, is there sufficient evidence to conclude that the average hospital stay for men is longer than the average hospital stay for women?

	Men	Women
Sample size	32	30
Sample mean	5.5 days	4.2 days
Population standard deviation	1.2 days	1.5 days

Step#1

H_0 :

H_1 :

Step#2

Level of significance: α

Step#3

Test Statistics

$$z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Step#4

Critical region

$$Z > Z_{\alpha}$$

$$Z < -Z_{\alpha}$$

$$Z > Z_{\alpha/2} \text{ or } Z < -Z_{\alpha/2}$$

If critical region satisfies reject H_0

Step#5

Calculation:

$$\bar{X}_1 =, \bar{X}_2 =, \sigma_1^2 =, \sigma_2^2 =, n_1 =, n_2 =$$

$$z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Step#6

Conclusion: